NPDE Note

1 Definitions, Classification of PDEs

Big $\mathcal{O}$: $f(t) = \mathcal{O}(g(t))$ as $t \to t_0$ if $\exists \varepsilon > 0, C > 0$ **s.t.** $\forall |t - t_0| < \varepsilon, |f(t)| \le C|g(t)|$

$\Leftrightarrow \lim_{t \to t_0} [f(t)/g(t)]$ bdd   ps: If not say $t \to t_0$, assume $t \to 0$

Classification: For $au_{xx} + bu_{xy} + cu_{yy} + du_x + eu_y + fu + g = 0$ $(a, b, c) \neq (0, 0, 0)$

1. **Hyperbolic**: $b^2 - 4ac > 0$ Admits non-trivial plane wave solutions
2. **Parabolic**: $b^2 - 4ac = 0$ (对变量 x, y 进行缩放后不会改变 PDE 类型)
3. **Elliptic**: $b^2 - 4ac < 0$ No non-trivial plane wave solutions

Weighted 2-Norm: $\|u\|_{2,\Delta x} = \sqrt{(\Delta x)^d}\|u\|_2$ where d s.t. $u : \mathbb{R}^d \to \mathbb{R}^n$ ↓不确定是否要保留, 应该要证明

For $d = 1$ and $\Delta x = \frac{D}{N}$, then: $\forall i, u_i = \mathcal{O}((\Delta x)^p) \Rightarrow \|u\|_{2,\Delta x} = \mathcal{O}((\Delta x)^p)$

For A symmetric: $\|A^{-1}\mathbf{w}\|_{2,\Delta x} \le \frac{1}{|\lambda_1|}\|\mathbf{w}\|_{2,\Delta x}$ $0 < |\lambda_1| \le \cdots \le |\lambda_n|$

Linear Algebra: **Positive Definite**: $\lambda_i > 0, \forall i$ **Negative Definite**: $\lambda_i < 0, \forall i$

Spectral Theorem: If A **symmetric**, then $A = Q\Lambda Q^\top$ where Q orthogonal, Λ diagonal with eigenvalues.

2 Global/Local Error, Convergence, Consistency, Stability

Exact & Approx. Sol: U_n: **approx. n-th time-step sol**; u_n: **exact sol at n-th time-step**.

Global Error $(e_n, \mathbf{e}(\Delta t))$: $e_n = U_n - u(t = n\Delta t)$ $\mathbf{e}(\Delta t) = [e_0, e_1, \dots, e_N]^\top$

Convergence: A numerical method is convergent if $\lim_{\Delta t \to 0} \|\mathbf{e}(\Delta t)\| = 0$

If $\lim_{\Delta t \to 0} \frac{\|\mathbf{e}(\Delta t)\|}{(\Delta t)^p} = C$ (i.e. $\lim_{\Delta t \to 0} \|\mathbf{e}(\Delta t)\| = \mathcal{O}((\Delta t)^p)$) $\Rightarrow$ **p-th order accurate**

Local Truncation Error (τ_n): 数值分析的离散迭代公式, 将 U_n 替换为 u_n 后的误差 *(LHS - RHS)*.

- e.g. For $\frac{U_{n+1} - U_n}{\Delta t} = f(U_n)$, $\tau_{n+1} = \frac{u_{n+1} - u_n}{\Delta t} - f(u_n)$ $\tau_0 = 0$
- e.g. 我们可以把线性数值方法写成 $Au = b$, A matrix, $u = [U_1, \dots, U_{N-1}]^\top$.
- Then: $Au_{\text{exact}} - b = \tau$ where $u_{\text{exact}} = [u_1, \dots, u_{N-1}]^\top$, $\tau = [\tau_1, \dots, \tau_{N-1}]^\top$

ps: $Ae = -\tau$ **where** $e = u - u_{\text{exact}}$ Define $\tau(\Delta t) = [\tau_1, \dots, \tau_{N-1}]^\top$

Consistency: If $\lim_{\Delta t \to 0} \|\tau(\Delta t)\| = 0$, the method is consistent.

Linear Equation: Goal: Finally change ODE/PDE into $Au = b$ where A **is matrix**, $u = [U_1, \dots, U_{N-1}]^\top$

Stability: A numerical method is stable if $\exists \Delta x_0$ s.t. For the Linear System $Au = b$:

1. $A(\Delta x)$ is invertible $\forall \Delta x < \Delta x_0$
2. $\|A^{-1}(\Delta x)\mathbf{w}\| \le C\|\mathbf{w}\|$ for all $\mathbf{w} \in \mathbb{R}^{N-1}$ (dim N depends on Δx but value of C does not)

Property: Consistent + Stable $\Rightarrow$ Convergent   and $\|\mathbf{e}(\Delta t)\| = \mathcal{O}(\|\tau(\Delta t)\|)$

3 Intro & Method of Lines

Taylor Series: $f(t + \Delta t) = f(t) + f'(t)\Delta t + \frac{1}{2}f''(t)(\Delta t)^2 + \cdots + \mathcal{O}((\Delta t)^N)$ **ps:** $u_{n+1} = u(n\Delta t + \Delta t)$

$$f(t) = f(a) + f'(a)(t - a) + \frac{f''(a)}{2!}(t - a)^2 + \cdots + \frac{f^{(N-1)}(a)}{(N-1)!}(t - a)^N + \mathcal{O}((t - a)^N)$$

Euler Methods: Forward: $\frac{du}{dt}|_{n\Delta t} \approx \frac{U_{n+1} - U_n}{\Delta t}$;   Backward: $\frac{du}{dt}|_{n\Delta t} \approx \frac{U_n - U_{n-1}}{\Delta t}$

Leapfrog: $\frac{du}{dt}|_{n\Delta t} \approx \frac{U_{n+1} - U_{n-1}}{2\Delta t}$ **Trapezoidal:** $\frac{1}{2}[\frac{du}{dt}|_{n\Delta t} + \frac{du}{dt}|_{(n+1)\Delta t}] \approx \frac{U_{n+1} - U_n}{\Delta t}$

Centred Difference: $\frac{d^2u}{dt^2}|_{n\Delta t} \approx \frac{U_{n-1} - 2U_n + U_{n+1}}{(\Delta t)^2}$

Method of Lines: Using $U_m(t) \approx u(a + m\Delta x, t)$, $U_m^n \approx u(a + m\Delta x, n\Delta t)$ a 为空间起点, 不重要

1. **For 1D: a.** 先对空间 x 离散化, 使用 U_m 表达,   此时 $\frac{d\mathbf{U}}{dt} = \dots$ where $\mathbf{U} = [U_1(t), U_2(t), \dots, U_{M-1}(t)]^\top$
   b. 对时间 t 使用 ODE 离散化, 使用 U_m^n 表达, 得到 Linear Equation.
2. Example: Crank-Nicolson for Heat Eq. $u_t = \kappa u_{xx}$: 对 x 用 centred difference; 对 t 用 trapezoidal.

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5 Appendix

Prototype PDEs: Laplace Eq.: $u_{xx} + u_{yy} = 0$ (E)   Poisson Eq.: $u_{xx} + u_{yy} = f(x, y)$ (E)

Helmholtz Eq.: $u_{xx} + u_{yy} + \alpha u = 0, \alpha \neq 0$ (E)   Heat Eq.: $u_t = \kappa(u_{xx} + u_{yy}), \kappa > 0$ (P)

Wave Eq.: $u_{xx} - u_{yy} = f(x, y)$ (H)   Advection Eq.: $u_t + vy_x + wu_y = 0$ (H)